## Remember

$F = C - P + 2$ Gibbs phase rule

Osmotic pressure $RT \ln \gamma_1 x_1 = - \underline{V}_1 \Delta P$

**Bubbles and Droplets** (boiling and clouds)

Similar analysis

**Droplets**

For simplicity, we will consider only the case of pure liquid and pure vapour.

The Gibbs phase rule suggests that:

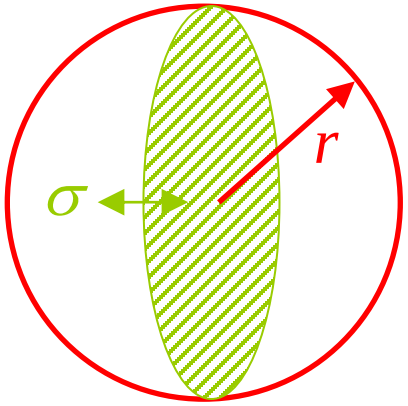
$$F = C - P + 2 = 1 - 2 + 2 = 1$$

i.e. a single boiling temperature for a pure liquid at given pressure

Again this is the wrong answer. Why?

We now have an additional variable – the droplet radius. So in fact we have two free parameters, for example the pressure and the droplet radius.

We have lifted the condition that the pressure in the liquid is equal to that in the vapour. This is because of surface tension:




surface tension [$Nm^{-1}$]

$$\left(P^{IN} - P^{OUT}\right)\pi r^2 = 2\pi r\sigma$$

projected force due to pressure difference

cohesive force across a hoop

$$P^{IN} - P^{OUT} = \frac{2\sigma}{r}$$

Pressure difference between the inside and outside of the droplet is a function of radius

It is this which has given us the extra degree of freedom

Different size droplets will have different boiling temperatures

What vapour pressure, $P^V$ is at equilibrium with a droplet of radius, $r$ at temperature, $T$?

Chemical equilibrium requires:

$$\mu^L = \mu^V$$

The liquid is now at higher $P$ than the vapour and so the chemical equilibrium is different to the saturated condition – this is analogous to the Poynting correction

$L$ being the droplet and $V$ the surrounding vapour (both are pure)

$$\mu^L = \underline{G}^L = \underline{G}^{SAT}(T) + \int_{P^{SAT}}^{P^L} \underline{V}^L dP \approx \underline{G}^{SAT}(T) + \underline{V}^L\left(P^L - P^{SAT}\right) \qquad \underline{V}^L \neq \text{fn}(P)$$

Saturation condition for pure vapour-liquid equilibrium

$T$,$P^{SAT}$, flat interface: $\underline{G}^L = \underline{G}^V$ by definition

$P^{SAT}$ is the saturated vapour pressure of the pure substance at $T$

$$\mu^V = \underline{G}^V = \underline{G}^{SAT}(T) + \int_{P^{SAT}}^{P^V} \underline{V}^V dP \approx \underline{G}^{SAT}(T) + RT \ln \frac{P^V}{P^{SAT}}$$

assuming vapour is IG

And so $\underline{V}^L\left(P^L - P^{SAT}\right) = RT\ln\frac{P^V}{P^{SAT}}$

same as Poynting correction, but now know $P^L - P^V = \frac{2\sigma}{r}$

$$\underline{V}^L\left(\frac{2\sigma}{r} + \cancel{P^V - P^{SAT}}\right) = RT\ln\frac{P^V}{P^{SAT}}$$

often assume this is small – see isotherm plot shortly

i.e. smaller droplets have a higher chemical potential and so are in equilibrium with a higher vapour pressure

Hence it is possible to have a meta-stable sub-cooled vapour

Smaller than equilibrium droplets shrink,
whilst larger than equilibrium ones grow

Alternatively, we can derive the same expression graphically by plotting an isotherm at $T$:

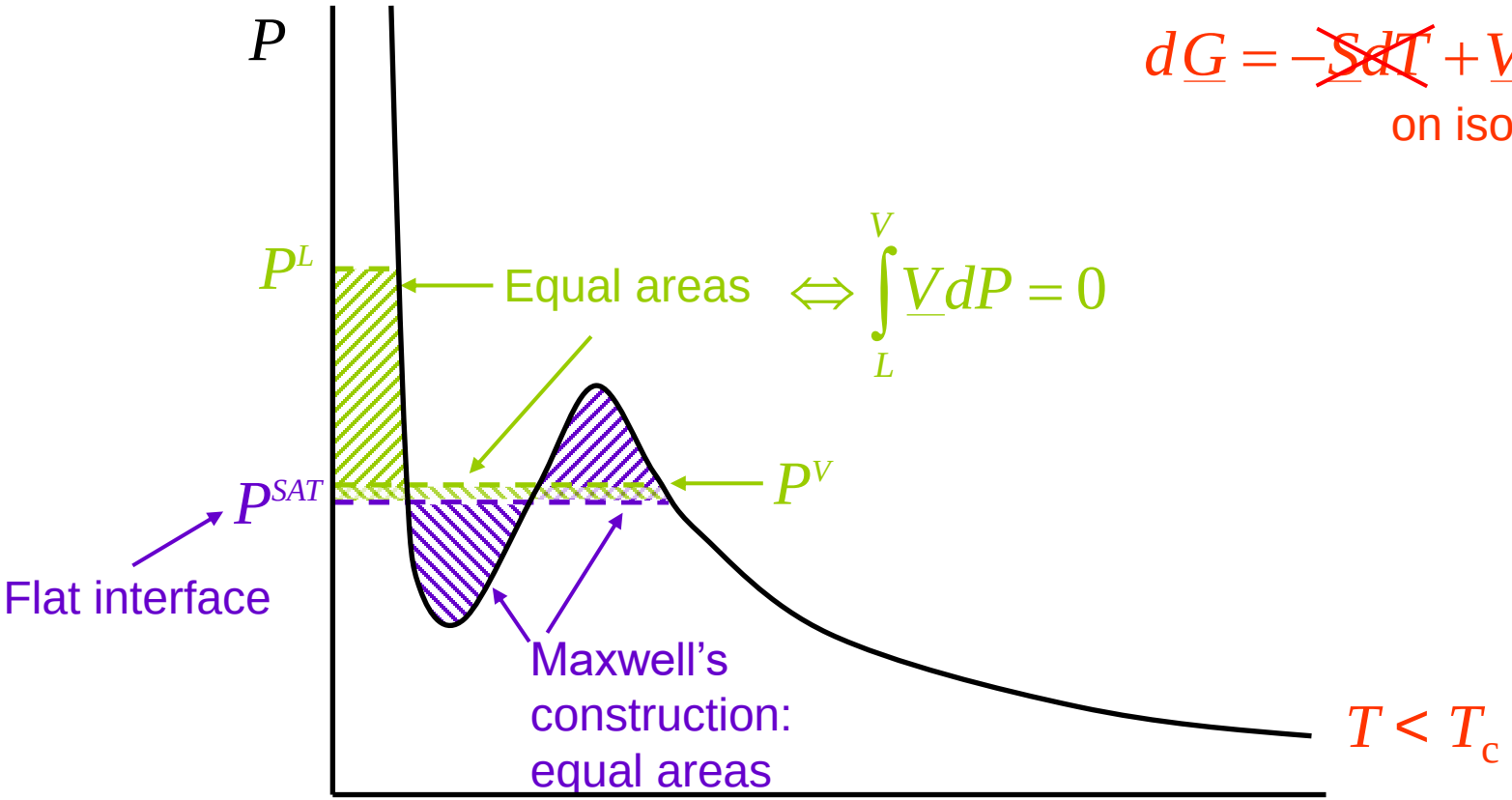
Purple = flat interface

Green = droplet

Chemical equilibrium $\underline{G}^V = \underline{G}^L$

$d\underline{G} = -\cancel{\underline{S}dT} + \underline{V}dP$

on isotherm



**Bubbles**

The analysis of bubbles is very similar to that of droplets

– again we have two degrees of freedom (atmospheric pressure and bubble radius for example), when the Gibbs phase rule (naively applied) predicts only one

At what temperature is a bubble of radius, $r$, just stable when the external pressure is atmospheric?

We will consider pure liquid

If the liquid is pure, then the vapour inside bubbles within that liquid must also be pure

The atmospheric gas need not be pure

Mechanical equilibrium requires (see above):

$$P^V = P^L + \frac{2\sigma}{r} = P^{ATM} + \frac{2\sigma}{r}$$ ← surface tension

$$\mu^V = \underline{G}^V = \underline{G}^{SAT}(T) + \int_{P^{SAT}}^{P^V} \underline{V}^V dP \approx \underline{G}^{SAT}(T) + RT \ln \frac{P^V}{P^{SAT}}$$ as before

$T, P^{SAT}$, flat interface: $\underline{G}^L = \underline{G}^V$ by definition (as for droplets)

and $\mu^L = \underline{G}^L \approx \underline{G}^{SAT}(T)$

$T, P$

$\underline{G}^L \neq \text{fn}(P)$ because $\underline{V}^L$ is small and we know that $P^L$ is not v. large 'cos it is $P^{ATM}$ (unlike droplet case)

So $$\frac{2\sigma}{r} = P^{SAT} - P^{ATM}$$

Now we know that at $T_b$ ,

$$P^{SAT}(T_b) = P^{ATM}$$

and Clausius-Clapeyron gives

assumed not fn($T$)

$$\frac{d \ln P^{SAT}}{dT} = \frac{\Delta \underline{H}^{VAP}}{RT^2} \Rightarrow \ln \frac{P^{SAT}(T)}{P^{ATM}} = \frac{\Delta \underline{H}^{VAP}}{R}\left(\frac{1}{T_b} - \frac{1}{T}\right)$$

Hence we can calculate the superheat required for a bubble to be at equilibrium at atmospheric pressure

The smaller the bubble the greater the superheat required

This results in superheated liquids being stable in the absence of nucleation sites

The thermodynamics of cooking pasta … consider adding salt to a pan of almost boiling water …



https://www.youtube.com/watch?v=2FcwRYfUBLM



## Boiling

occurs when atmospheric pressure is equal to vapour pressure

Superheat may be necessary for the kinetic formation of bubbles

If only pure vapour is present, we are automatically at the boiling point whenever VLE is achieved. However, we are used to dealing with liquids in air and in this case VLE is possible without boiling

Boiling point for a liquid in air is a function of atmospheric pressure (think about a pressure cooker) and so altitude.

Just as freezing point is depressed by the presence of an impurity, so boiling point is elevated by the presence of an involatile impurity (e.g. salt or sugar).

[Exercise – derive an expression for
elevation of boiling point]

This is usefully thought of graphically:

